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RELATIONS.

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POLARITY SENSITIVITY AS INHERENT SCOPE RELATIONS

by

WILLIAM ALLEN LADUSAW, B.A., M.A.

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POLARITY SENSITIVITY AS INHERENT SCOPE RELATIONS

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DEDICATION

To my parents

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TABLE OF CONTENTS

	<u>Page</u>
ACKNOWLEDGEMENTS	iv
CHAPTER I: Introduction	1
Footnotes	14
CHAPTER II: A Semantic Theory for English	
0. Some Assumptions about the Semantic Component of a Grammar	15
1. The Relation of Semantics to Syntax	16
2. A Translation-type Semantic Component for English	19
3. Interpretation v. Reading	32
4. Karttunen-Peters Semantics	35
Footnotes	44
CHAPTER III: The Definition of Scope	
0. Definitions of Scope for a Formal Language	47
1. A Geometric Definition on the Syntax of English	51
2. A Geometric Definition on the Syntax of "Logical Form"	54
3. A Non-Geometric Definition	58
4. A Type-Theory for English	71
5. The Scope of Auxiliary <u>Not</u>	78
Footnotes	85
CHAPTER IV: <u>Some</u> , <u>Any</u> , and Inherent Scope	
0. Polarity Sensitivity as Inherent Scope	90
1. The Problem of <u>Any</u> and <u>Some</u>	91
2. Polarity Sensitive <u>Any</u> is an Existential Determiner	94
2.1 Arguments Against <u>Any</u> 's Having Wide Scope	96

	<u>Page</u>
2.2 Arguments from the Formulation of Meanings	101
3. Free-Choice <u>Any</u>	104
4. Modifying the Interpretation Procedure to Represent Inherent Scope	106
Footnotes	114
CHAPTER V: Polarity Sensitive Sentence Adverbs	
0. Introduction	116
1. Assertional Adverbs of Quantification	117
2. Implicative Adverbs of Aspect	120
3. Some Additional Considerations About <u>Yet</u> and <u>Already</u>	124
4. Positive <u>Anymore</u>	127
5. Modification of the Fragment	128
CHAPTER VI: Triggers and Anti-Triggers	
0. Introduction	131
1. Reduction of All Triggers to Negation	134
2. A Scale-Entailment Approach to NPI Triggering	137
3. Scale Reversal and Downward-Entailing	143
4.0 Downward-Entailing as [+Affective]	147
4.1 Determiners and Quantifier NP's	147
4.2 Sentence Adverbs	153
4.3 $\bar{S}$ -Complement Verbs	157
4.4 $\bar{VP}$ -Complement Verbs	159
4.5 <u>Too</u> and <u>Enough</u>	161
4.6 Prepositions	162
4.7 <u>Only</u>	165
4.8 Adverbial Conjunctions	166
4.9 Conclusion	168

	<u>Page</u>
5. Anti-Triggers	168
6. Summary	171
Footnotes	172
CHAPTER VII: "Double Negatives"	174
Footnotes	185
CHAPTER VIII: Polarity Sensitive Items in Questions . . .	186
Footnotes	194
CHAPTER IX: Summary and Conclusion	
0. The Semantic Nature of Polarity Filtering	195
1. Island Constraints	203
2. The Left-Right Order Restriction	206
3. What Do N and W Represent?	207
Footnotes	211
BIBLIOGRAPHY	212
VITA	216

CHAPTER I: INTRODUCTION

The goal of this investigation was to establish the nature of the principles of English grammar which account for the varying acceptability of the sentences like those in (1-4) which contain so-called polarity sensitive lexical items (PSI's) like any, yet, already, and some. These and many other lexical items and some near-idiomatic expressions in English have a distribution in sentences which is not completely predictable from their syntactic properties. The assignment of any to the category of Determiners, and yet to a subcategory of adverbs in order to generate sentences

- (1) *John has talked to any of the students
- (2) John hasn't talked to any of the students
- (3) John has already talked to some of the students
- (4) ?John hasn't already talked to some of the students

like (2) will incorrectly predict their acceptable appearance in (1) unless some additional restriction is placed on them. This dissertation formulates and defends the thesis that the principles which account for the limited distribution of PSI's are semantic and not syntactic. The large linguistic literature on the topic of polarity sensitivity includes proposals that the filtering of (1) and (4) should be expressed by a syntactic transformation (Klima 1964), by a semantic "Interpretive Rule" which alters features of lexical items during a derivation of a "semantic representation" (Jackendoff 1969), by a semantic generalization based upon the notion of entailment (Baker 1970b), and by a generalization which involves pragmatic principles using scalar implications (Fauconnier 1975).

The hypothesis that the anomaly of (1) and (4) is to be explained by semantic filtering has been an attractive one, but one whose formulation and confirmation have been hampered by the lack

of a semantic theory general enough to cover the wide range of relevant data. Negative-polarity items like any and yet have been considered to be subject to a semantic restriction which requires that they be interpreted within the semantic scope of a negation (either overt or "covert"). In order for this hypothesis to have testable consequences, it must be considered in the context of some semantic theory which provides formulations of the notions of scope and negation. I have embedded this inherent scope hypothesis for PSI's within a theory of grammar which incorporates a model-theoretic semantic component which interprets surface structures of English directly, and argue that it is in fact a very good account of a large part of the problem posed by polarity sensitivity, though it is not yet a complete theory of it.

The results of this investigation are as destructive as they are constructive. Before a complete theory of polarity sensitivity can be developed, several past hypotheses and assumptions must be discarded. Foremost among these is the idea that (1) and (4) are syntactically ill-formed in some way. I will argue below that the explanation for the unacceptability of (1) and (4) should be sought in semantic theory since the principles which predict what sentences will be "polarity violations" must be stated almost entirely in terms of semantic properties of sentences and their component expressions.

Other incorrect assumptions are embedded subtly in the terminology traditionally used to describe the phenomena. Saying that certain items are "polarity sensitive", and that there are Affirmative-polarity items (API's) and Negative-polarity items (NPI's) suggests that PSI's are sensitive to whether a sentence is affirmative or negative, and that API's and NPI's show complementary co-occurrence restrictions to this single feature of the sentence. This view of polarity sensitivity turns out to be incorrect.

The property that NPI's are sensitive to is not a property of sentences, it is a property that only expressions with functional

meanings can have. (Sentence meanings are not functional--they do not take meanings of other expressions as arguments.) Certain expressions which have this property are said to "trigger" NPI's, because NPI's are acceptable only in sentences which contain one of these expressions. The terminology is a relic of the analysis that used a transformational rule governed by the class of triggers (Klima 1964) to mark NPI's as acceptable. I have preserved the term here for lack of a better one, and will show that whether an expression (lexical or phrasal) is a trigger is predictable from its meaning. NPI's are items which must be semantically interpreted as being in the semantic scope of an expression with an appropriate meaning. The relevant aspect of meaning is not a property of sentences and so it is impossible to state the distribution of NPI's in terms of a property of sentences.

API's on the other hand are more reasonably analyzed as expressions which are sensitive to whether the sentence is negative or affirmative. But we must make a distinction between sentences which are negative propositions (like (2)) and sentences which are the contradiction or denial of affirmative propositions (like (4)). API's may appear in sentences which are denials of affirmative assertions, though not in negative sentences. (4) is acceptable as a denial of a previous assertion or belief that John has talked to someone, but unacceptable as a straight assertion of the proposition expressed by (2).

This asymmetry in the sensitivity of NPI's and API's is obscured in every present account of polarity sensitivity. Below we will develop a way of stating a restriction which requires NPI's to be in the scope of a trigger (a *downward-entailing expression*) and which prevents API's from falling in the scope of a negation unless it is a negation which is used to deny or contradict some assertion or belief. Only by noting this asymmetry between the sensitivity of API's and NPI's can we begin to account for sentences like (5) and (6) which contain clausemate NPI's and API's both, and whose

acceptability is proscribed by the traditional view of polarity sensitivity.

- (5) I doubt that anyone has already finished the exam
- (6) I'm surprised that someone hasn't already left
without anything to eat

One very difficult phenomenon to describe has been the effect that PSI's have on the uses to which questions containing them may be put. I will argue that the appearance of NPI's in questions is not a case of semantic triggering, and show that we can predict the "bias" of questions containing PSI's on the basis of the semantic properties associated with NPI's and API's, and hence associated with the meanings of questions containing them.

Finally, we must remember that not all NPI's are alike. The account given here works for the largest class of NPI's: any, ever, yet, and what Schmerling 1971 argues is an open class of items like lift a finger, drink a drop, eat a bite, budge an inch, etc. Other NPI's like either, until, and the modal need do not admit of this treatment. By demonstrating here the principles which govern the distribution of the largest class of NPI's, it will be possible to move on to analyze the distinctions which previous terminology has obscured.

Since Klima's (1964) analysis of negation in English first represented some and any as alternative phonological realizations of one indeterminate quantifier, several approaches have been made to the general problem of how to account for the restricted distribution of PSI's. These approaches have all concentrated on finding some appropriate level of grammatical description at which a filtering condition might be stated which would restrict their distribution. This filtering condition must eliminate two kinds of "polarity violations": NPI's in sentences which do not also contain a "trigger" (e.g. (1)) and API's in sentences which also

contain what I will call an "antitrigger", such as a negation (e.g. (4)).

Klima stated this filtering condition as the transformational rule of Indefinite Incorporation. Since the filtering condition was a syntactic rule, (1) was considered to be syntactically ill-formed. Jackendoff 1969 reanalyzed Klima's syntactic filtering condition within an "Interpretive Semantics" framework, arguing that (1) should be marked as semantically anomalous under conditions which are basically the same as Klima's. Baker 1970b discussed the problem of extending these analyses to complex sentences which he called "double negatives". His filtering condition made use of the semantic notion of entailment to account for the occurrence of (1) and (4) as sub-constituents of the acceptable (7a) and (7b) respectively.

- (7)a. I'm surprised that John has talked to any of
the students yet
- b. I'm surprised that John hasn't already talked
to some of the students

Finally, Fauconnier 1975 suggested that the parallels between "syntactic polarity items" like yet, any, and ever and other more "semantic" or "pragmatic" polarity items like even-phrases and what he called quantificational superlatives show that a complete theory of polarity sensitivity must take into account what he called "pragmatic scales" and non-logical but pragmatically reasonable entailments.

Klima's original analysis of the near-complementary distribution of API's and NPI's like some and any, already and yet, and once and ever, made NPI's derived lexical forms rather than basic lexical items which might be inserted into deep structures. Underlying each of these pairs of an API and an NPI was an abstract lexical item carrying the feature [+Indeterminate]. Possible NPI trigger items like not, no, and seldom were marked [+Affective]. A transformational rule called "Indefinite Incorporation" optionally

marked [+Indeterminate] lexical items [+Indefinite] if they stood in a certain structural relation (*in construction with*¹) a [+Affective] lexical item. These abstract lexical items took a special form (the NPI) if they were [+Indefinite]. Otherwise they were realized as the API. Since NPI's were derived forms, they would successfully be restricted to appearing only in construction with triggers. Since Indefinite Incorporation was optional, API's might also appear in construction with a trigger.

This analysis was revised and extended by Jackendoff 1969 to eliminate certain undesirable formal and empirical consequences. Jackendoff objected to the use of syntactic rules and features to derive the lexical forms of the NPI's and API's from abstract lexical items, particularly when there was so little morphological similarity between the two realizations of Klima's indeterminates. Further, Klima's suppletion approach would not generalize naturally to provide an account of all PSI's, since most do not come in pairs. To make the approach general, anymore and at all (for example) would have to be analyzed as indefinite forms of an indeterminate lexical item which would be phonologically unrealizable when not in construction with a [+Affective] trigger.

Jackendoff treated each PSI as an independent lexical item which was "semantically" marked with an inherent value for [$\pm$ Indeterminate] and could be marked with a value for a feature [$\pm$ X] as well. API's were inherently [+X]; NPI's were inherently [-X]. (Jackendoff used [X] as the name of the feature only to facilitate his discussion of its relation to and possible reduction to other syntactic and "semantic" features.) To accomplish the filtering² of (1) and (4), he proposed the rule of "semantic interpretation" in (8a) (=Jackendoff's (58)) with its convention for applicability, (8b).

- (8)a. [+Indeterminate] + [-X]/in construction with
Affective
[+X] elsewhere

(8) *(continued)*

- b. If an indeterminate is unspecified with respect to X, the rule fills in the feature according to the environment. If the indeterminate is already marked with respect to X, the sentence is marked semantically anomalous if the inherent feature and the feature assigned by the rule disagree.

This reconstruction of Klima's analysis maintains the structural characterization of the relation between NPI's and their triggers which Indefinite Incorporation expressed because it still uses the structural notion in-construction-with. Rather than mapping Phrase Markers onto Phrase Markers as a syntactic transformation would, (8a) helps to develop Phrase Markers into a kind of structured object which Jackendoff identifies as semantic. (8) predicts that NPI's will always appear in construction with affective items and that API's never will. If we assume that this rule can be made optional in certain cases³, it correctly predicts the judgments in (1-4) and is a candidate for extension to more complex cases.

Baker 1970b introduces into consideration complex sentences which rules like (8) in principle cannot account for. His "double negative" sentences like those in (9) contain embedded sentences which are polarity violations like (1d): they contain API's in construction with not. Baker considers a possible revision of a

- (9)a. There isn't anyone in this camp who wouldn't rather be in Montpelier
- b. I find it impossible to believe that someone else couldn't do a far better job than our present governor
- c. You can't convince me that someone isn't still holed up in that cave

rule like (8) which would allow some embedding contexts to cancel the anti-triggering effect of a negation in an embedding clause, in effect making a superficially negated clause count as semantically

affirmative for the purposes of the rules stating the distribution of PSI's. Baker later rejects this "polarity reversal rule" in favor of a generalization which attempts to predict the acceptable contexts for both NPI's and API's in two stages. The first stage is a generalization using a semantic notion of scope rather than a syntactic relation like in-construction-with. (10) (p. 179) sets

- (10) Negative-polarity items are appropriate in structures within the scope of negations whereas affirmative-polarity items are appropriate elsewhere.

up an initial complementarity in the distribution of API's and NPI's. He attempts to account for the apparent exceptions to (10) posed by "double negatives" and triggering contexts which are "pseudo-negatives" like doubt and be-surprised by the principle in (11).

- (11) Given semantic representations P_1 and P_2 satisfying the following conditions:

- (A) $P_1 = X_1 \text{ } \forall \text{ } Z_1$ and $P_2 = X_2 \text{ } \forall \text{ } Z_2$, where $\forall$ is itself a well-formed semantic representation;
 (B) P_1 entails P_2 ;

then the lexical representation appropriate to $\forall$ in P_2 [by (10)] is also appropriate to $\forall$ in P_1 .

There is a reason, in addition to the one that Baker gives, for rejecting any attempt like Jackendoff's or Baker's suggested revision of it which accounts for the sentences Baker discusses with features whose values are changed during the course of a derivation by the presence of various trigger elements. This reason also provides evidence that even Baker's semantic generalization is insufficient to account for all of the cases that we would want a general theory of polarity sensitivity to account for.

Jackendoff's proposal, Baker's suggested revision of it, and Baker's own proposal all have one feature in common: they assume that NPI's must stand in some relation to a negation (or trigger)

and that API's may not stand in that relation to a negation. That is, that (1) and (4) are a natural class of cases: polarity violations. While embedded occurrences of violations like (1) may always be accounted for by revising one's notion of Affective or the relation that must hold between a trigger and a NPI, violations like (4) require some grounds for treating an overtly negated clause as affirmative (as in (7b)) by somehow cancelling the effect of the negation.

The polarity reversal approach discussed by Baker which flips feature values with an alpha-rule makes one prediction which is easily falsified. The environment for changing an $[\alpha\text{polarity}]$ embedded clause to a $[-\alpha\text{polarity}]$ clause will always be the same, whether the value of α is + or -. By including an additional negation in the embedded clause, the result of applying the rule will be reversed for that clause. Hence clauses like (1) and (4) should have identical distributions. If some context can reverse the polarity of an embedded clause (as in (7b)) to be affirmative, enabling an API to occur with a clausemate negation, it will also reverse it to enable an NPI to occur without a negation (as in (7a)). So $\bar{S}$ -embedding verbs should fall into two classes: reversing and non-reversing. They should allow either only clauses like (2) and (3), or allow any of the sentences in (1-4) as complements.

There are, however, verbs which will allow all but (1) as their complements as (12-14) show.

- (12)a. *John found out that Bill has talked to anyone
 yet
 b. John found out that Bill hasn't talked to anyone
 yet
 c. John found out that Bill has already talked to
 someone
 d. John found out that Bill hasn't already talked to
 someone

- (13)a. *I hope that Bill has talked to anyone yet
- b. I hope that Bill hasn't talked to anyone yet
- c. I hope that Bill has already talked to someone
- d. I hope that Bill hasn't already talked to someone
- (14)a. *I'm glad that Bill has talked to anyone yet
- b. I'm glad that Bill hasn't talked to anyone yet
- c. I'm glad that Bill has already talked to someone
- d. I'm glad that Bill hasn't already talked to
 someone

This suggests that however we account for double negatives, we should not treat (1) and (4) as a natural class. There is an obvious asymmetry between the degree of unacceptability of (1) and (4) which is obscured if they are considered to be violations of the same type. (1) is hopelessly unsalvageable as a complete utterance, even when surrounded by a discourse. (4) on the other hand is perfectly acceptable (as Baker notes in his footnote 2) "as an emphatic denial of a preceding speaker's assertion." The (d) sentences in (12-14) seem acceptable, but their use would suggest that it had been the expectation or belief of the person referenced by the subject of the sentence that the affirmative form of their complement was true.

The sentences in (12-14d) are counterexamples to Baker's proposal as well. In order for (11) to license them as acceptable, they must entail some sentence with the proposition "Bill has already talked to someone" as a coherent part of its meaning. I have been unable to find any entailment of (12-14d) which could do this. The closest candidates seem at most conversational implicatures of these sentences, expressed by the sentences of (15).

- (15)a. John used to think Bill had already talked to
 someone
- b. I fear that Bill has already talked to someone
- c. I would have been sorry if Bill had already
 talked to someone

Baker 1970b and the revision of Jackendoff 1969 in Jackendoff 1972 place nearly the entire burden of polarity filtering on the semantic component of the grammar. They require us to be able to

state what I will call an inherent scope filter for PSI's. Baker's inherent scope statement is the first stage of his two-stage generalization in (10), repeated here as (16a). It suggests a correlation between polarity sensitivity and inherent scope with respect to negation: NPI's must be in the scope of negations, API's must not. Jackendoff's inherent scope statement in (16b) is stated only for

- (16)a. NPI are appropriate in structures within the scope of negations, whereas API are appropriate elsewhere.
- b. (Jackendoff 1972, p. 337) Indefinites may occur only when dependent on "certain modal operators, the affectives."

NPI's (which he calls indefinites, following Klima) but uses the cover term "affective" for the class of triggers, rather than making negations the only triggers.

As the title suggests, the majority of this dissertation is concerned with making the hypothesis suggested by the statements in (16) more explicit within a theory of the syntax and semantics of English which is both explicit and complete enough to properly test it as a general hypothesis about the treatment of all PSI's.

In Chapter II I lay out a theory of the semantic component of English which uses the translation theory of Montague 1970 as modified by Cooper 1975 to give a model-theoretic interpretation to sentences based on their shallow syntactic structure. This semantic theory provides the basis for a general definition of semantic scope in Chapter III, as well as an interpretation for the initial fragment of English given there.

In Chapter IV I turn to the analysis of any and some and argue that we should attribute their differing polarity sensitivity not to differences in their syntax or their truth-conditional or implicative meaning, but to independent attributes of their meaning. These attributes are interpreted as the mark of inherent scope with respect to negation: for NPI's, it is obligatory narrow scope, and

for API's, it is a preference for wide scope with respect to negation.

The filtering condition which accounts for the distribution of any and some is used to predict the distribution of polarity sensitive sentence adverbs in Chapter V. By including the category of sentence adverbs in our fragment, we give analyses of both new triggering lexical items (e.g. never, rarely, and seldom), NPI's (e.g. ever, anymore, and yet) and API's (already, still, sometimes).

In Chapter VI I turn to the problem of predicting what words may serve as triggers for NPI's and to the question of whether there is a distinction in fact between triggers and what I have called antitrigger, the items in whose scope the API's may not appear. To replace Klima's feature [+Affective] I define the semantic property of being a *downward-entailing expression* which is very successful as a predictor of triggerhood, and discuss its relation to Fauconnier's analysis of polarity sensitivity using "scales".

Chapter VII examines the evidence for Baker's second stage generalization as an account of "double negatives" and attempts to treat the majority of them as embedded occurrences of negated sentences which would be denials as matrix sentences. Chapter VIII discusses why questions are polarity neutral environments and the effects of PSI's on the potential uses for questions which contain them.

In Chapter IX I review the argument and conclude that the success of interpreting polarity sensitivity as a sign of inherent scope preference merits the extension of it to other PSI's. I discuss some of the problems remaining for further research, such as the varying acceptability of NPI's in different triggering contexts, and the fact that some NPI's do not seem to be amenable to the treatment given here for any, ever, yet, etc. I also discuss two

possible interpretations for the attributes of meaning which I introduce to describe the polarity sensitivity of these items, and the issues that bear on deciding between them.

FOOTNOTES TO CHAPTER I

¹ α is in construction with β iff the node which immediately dominates β also dominates α .

² Jackendoff says (p. 233) that the rule in (8) should not be considered to have merely a filtering function since for indeterminates such as NP's with a as determiner which are unmarked for [X], the value of [X] is set by the rule and hence the rule also contributes to the construction of the "reading" of a sentence to which it has applied. Since the notions of semantic interpretation and reading that we will be using are so different from his, this point is irrelevant to this discussion.

³ For example, questions and the antecedents of conditionals allow both [+X] and [-X] constituents, and we must be able to derive sentences like (i) where an API (someone) is in the syntactic scope of a negation, but not in its semantic scope.

(i) John didn't talk to someone

⁴ Jackendoff's *dependent on* is the semantic notion of *be in the scope of*. He reserves the term *scope* for a syntactic relation and relates the syntactic and semantic relation so that anything which is in the syntactic scope of an operator is optionally dependent on it. For further discussion, see Chapter III.

CHAPTER II

A SEMANTIC THEORY FOR ENGLISH

0. Some Assumptions about the Semantic Component of a Grammar

My discussion of polarity sensitivity will proceed as I justify a syntactic and semantic analysis of a fragment of English large enough to contain an interesting range of PSIs and trigger items and constructions. Because I want to claim that the results of this study are relevant to any approach to polarity sensitivity, I will from time to time discuss how the analysis might change given alternative assumptions. In this chapter, I will lay out the general approach to natural language semantics taken here and what its advantages are for this study.

I take the task of a semantic analysis of English to be to associate a non-linguistic object which we will construe as a meaning with each syntactically well-formed expression (of any category). Further, I assume that the grammar specifies the association in such a way that meanings are arbitrarily assigned only to a finite subset of the expressions of the language--its lexical items or basic expressions. The meanings of complex expressions are assigned as a function of the meanings of their component expressions and the syntactic operations which form them. Finally I assume that at least one component of the meaning to be assigned to sentences of the language is a representation of their truth-conditions in terms of a possible-worlds-type model-theoretic semantics. Hence part of the meaning associated with basic lexical expressions will be what they contribute to the truth-conditions of sentences which contain them.

These three assumptions, of course, are the basis for the Fregean semantic theories used by linguists interested in applying